WORKED SOLUTION

STEP II 1989, Question 9

Pure • Matrix Exponential Systems

STEP II 1989, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q9

The key idea

Everything hinges on one small calculation: $\mathbf{A}^2$ turns out to be the identity. Then every even power of $\mathbf{A}$ is $\mathbf{I}$ and every odd power is $\mathbf{A}$, so the series splits into the even and odd parts of the exponential series, which are $\cosh t$ and $\sinh t$. For the differential equation we copy the scalar integrating-factor method, with the matrix $\mathbf{F}$ playing the role of $\mathrm{e}^{\int a\,\mathrm{d}t}$.

Solution

The matrix $\mathbf{F}$

Multiplying out,

$$\mathbf{A}^2 = \begin{pmatrix} -3 & -1 \\ 8 & 3 \end{pmatrix}\begin{pmatrix} -3 & -1 \\ 8 & 3 \end{pmatrix} = \begin{pmatrix} 9-8 & 3-3 \\ -24+24 & -8+9 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}.$$

Hence $\mathbf{A}^{2k} = (\mathbf{A}^2)^k = \mathbf{I}$ and $\mathbf{A}^{2k+1} = \mathbf{A}^{2k}\mathbf{A} = \mathbf{A}$ for every $k \geq 0$. Writing $\mathbf{I} = \frac{t^0}{0!}\mathbf{A}^0$ and splitting the series into even and odd n (each entry of the matrix series is a rearrangement of absolutely convergent scalar series, so this is allowed),

$$\begin{aligned} \mathbf{F} &= \sum_{k=0}^{\infty} \frac{t^{2k}}{(2k)!}\mathbf{A}^{2k} + \sum_{k=0}^{\infty} \frac{t^{2k+1}}{(2k+1)!}\mathbf{A}^{2k+1} \\ &= \mathbf{I}\sum_{k=0}^{\infty} \frac{t^{2k}}{(2k)!} + \mathbf{A}\sum_{k=0}^{\infty} \frac{t^{2k+1}}{(2k+1)!} = \mathbf{I}\cosh t + \mathbf{A}\sinh t, \end{aligned}$$

using the Maclaurin series $\cosh t = \sum t^{2k}/(2k)!$ and $\sinh t = \sum t^{2k+1}/(2k+1)!$. This is the required form.

The inverse

Let $\mathbf{G} = \mathbf{I}\cosh t - \mathbf{A}\sinh t$. Since $\mathbf{I}$ and $\mathbf{A}$ commute, we can multiply out as with ordinary numbers:

$$\mathbf{FG} = \mathbf{I}\cosh^2 t - \mathbf{A}^2\sinh^2 t + (\mathbf{A} - \mathbf{A})\cosh t \sinh t = \mathbf{I}(\cosh^2 t - \sinh^2 t) = \mathbf{I},$$

using $\mathbf{A}^2 = \mathbf{I}$ and $\cosh^2 t - \sinh^2 t = 1$. The same computation gives $\mathbf{GF} = \mathbf{I}$, so $\mathbf{F}^{-1} = \mathbf{I}\cosh t - \mathbf{A}\sinh t$.

The derivative

Differentiating entry by entry (the matrices $\mathbf{I}$ and $\mathbf{A}$ are constant),

$$\frac{\mathrm{d}\mathbf{F}}{\mathrm{d}t} = \mathbf{I}\sinh t + \mathbf{A}\cosh t, \qquad \text{while} \qquad \mathbf{FA} = \mathbf{A}\cosh t + \mathbf{A}^2\sinh t = \mathbf{A}\cosh t + \mathbf{I}\sinh t.$$

These agree, so $\dfrac{\mathrm{d}\mathbf{F}}{\mathrm{d}t} = \mathbf{FA}$. Note also that $\mathbf{FA} = \mathbf{AF}$, since $\mathbf{F}$ is built from $\mathbf{I}$ and $\mathbf{A}$.

The differential equation

In the scalar equation $r' + ar = 0$ we multiply by e^{at}, whose derivative is $\mathrm{e}^{at}a$, so that the left-hand side becomes an exact derivative. The matrix $\mathbf{F}$ has exactly this property, $\mathbf{F}' = \mathbf{FA}$, so we multiply the equation on the *left* by $\mathbf{F}$:

$$\mathbf{F}\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} + \mathbf{FA}\mathbf{r} = \mathbf{0}.$$

By the product rule (which holds for matrix products, as long as we keep the order of the factors),

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{Fr}) = \frac{\mathrm{d}\mathbf{F}}{\mathrm{d}t}\mathbf{r} + \mathbf{F}\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \mathbf{FA}\mathbf{r} + \mathbf{F}\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t},$$

which is precisely the left-hand side. So the equation says

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{Fr}) = \mathbf{0},$$

and $\mathbf{Fr}$ is a constant vector $\mathbf{c}$.

> **Ask yourself** Why multiply on the left rather than the right, and how do we find $\mathbf{c}$?

The product $\mathbf{rF}$ doesn't even make sense ($\mathbf{r}$ is a 2×1 column), and the product rule produced $\mathbf{FAr}$, which matches the term $\mathbf{F}(\mathbf{Ar})$ only when $\mathbf{F}$ is on the left. For the constant, put $t = 0$: there $\cosh 0 = 1$ and $\sinh 0 = 0$, so $\mathbf{F} = \mathbf{I}$ and

$$\mathbf{c} = \mathbf{I}\,\mathbf{r}(0) = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}.$$

Now multiply $\mathbf{Fr} = \mathbf{c}$ on the left by $\mathbf{F}^{-1}$, which we found above:

$$\mathbf{r} = \mathbf{F}^{-1}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \cosh t\begin{pmatrix}\alpha\\ \beta\end{pmatrix} - \sinh t\begin{pmatrix}-3 & -1\\ 8 & 3\end{pmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \cosh t\begin{pmatrix}\alpha\\ \beta\end{pmatrix} - \sinh t\begin{pmatrix}-3\alpha - \beta\\ 8\alpha + 3\beta\end{pmatrix}.$$

Reading off the components,

$$x(t) = \alpha \cosh t + (3\alpha + \beta)\sinh t,$$
$$y(t) = \beta \cosh t - (8\alpha + 3\beta)\sinh t,$$

as required.

Checks

- At $t = 0$ the formulae give $x = \alpha$, $y = \beta$, as they should.
- Direct substitution: $x' = \alpha \sinh t + (3\alpha + \beta)\cosh t$, and $-(-3x - y) = 3x + y = (3\alpha + \beta)\cosh t + \alpha \sinh t$ (using $9\alpha + 3\beta - 8\alpha - 3\beta = \alpha$). So $x' = -(\text{first row of } \mathbf{Ar})$, and similarly for y.
- Numerically, summing the series to 60 terms agrees with $\mathbf{I}\cosh t + \mathbf{A}\sinh t$, and a Runge–Kutta solution of the system with $\alpha = 1$, $\beta = 0$ gives $x(2) \approx 14.6428$, matching $\cosh 2 + 3\sinh 2$.

Common slips

- Multiplying by $\mathbf{F}$ on the right, or writing $\mathbf{AF}$ and $\mathbf{FA}$ interchangeably without noting why that's allowed here (they commute because $\mathbf{F}$ is a combination of $\mathbf{I}$ and $\mathbf{A}$).
- Forgetting the $\mathbf{I}$ term (the $n = 0$ term) when splitting the series, which loses the $\cosh t$ constant term 1.
- Sign errors when multiplying $-\mathbf{A}$ into $(\alpha, \beta)^{\mathrm{T}}$.

Beyond STEP

The matrix $\mathbf{F}$ is the matrix exponential $\mathrm{e}^{t\mathbf{A}}$, and the method here shows that $\mathbf{r}' = \mathbf{Mr}$ always has solution $\mathbf{r} = \mathrm{e}^{t\mathbf{M}}\mathbf{r}(0)$; here $\mathbf{M} = -\mathbf{A}$, which is why the answer involves $\mathbf{F}^{-1} = \mathrm{e}^{-t\mathbf{A}}$. In general one computes $\mathrm{e}^{t\mathbf{M}}$ by diagonalising $\mathbf{M}$: our $\mathbf{A}$ has eigenvalues ± 1 (that is what $\mathbf{A}^2 = \mathbf{I}$ says), which is where $\mathrm{e}^{\pm t}$, and so cosh and sinh, come from. Strang's *Introduction to Linear Algebra* has a good chapter on matrix exponentials and systems of differential equations, and Boyce and DiPrima's *Elementary Differential Equations* treats linear systems in detail.

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